Adafruit_Fingerprint Class Reference

! Helper class to communicate with and keep state for fingerprint sensors More...

```
#include <Adafruit_Fingerprint.h>
```

Public Member Functions

	Adafruit_Fingerprint (HardwareSerial *hs, uint32_t password=0x0) Instantiates sensor with Hardware Serial. More...
	Adafruit_Fingerprint (Stream *serial, uint32_t password=0x0) Instantiates sensor with a stream for Serial. More...
void	**begin** (uint32_t baud) Initializes serial interface and baud rate. More...
boolean	**verifyPassword** (void) Verifies the sensors' access password (default password is 0x0000000). A good way to also check if the sensors is active and responding. More...
uint8_t	**getParameters** (void) Get the sensors parameters, fills in the member variables status_reg, system_id, capacity, security_level, device_addr, packet_len and baud_rate. More...
uint8_t	**getImage** (void) Ask the sensor to take an image of the finger pressed on surface. More...
uint8_t	**image2Tz** (uint8_t slot=1) Ask the sensor to convert image to feature template. More...
uint8_t	**createModel** (void) Ask the sensor to take two print feature template and create a model. More...
uint8_t	**emptyDatabase** (void) Ask the sensor to delete ALL models in memory. More...
uint8_t	**storeModel** (uint16_t id) Ask the sensor to store the calculated model for later matching. More...
uint8_t	**loadModel** (uint16_t id) Ask the sensor to load a fingerprint model from flash into buffer 1. More...
uint8_t	**getModel** (void) Ask the sensor to transfer 256-byte fingerprint template from the buffer to the UART. More...
uint8_t	**deleteModel** (uint16_t id) Ask the sensor to delete a model in memory. More...
uint8_t	**fingerFastSearch** (void) Ask the sensor to search the current slot 1 fingerprint features to match saved templates. The matching location is stored in **fingerID** and the matching confidence in **confidence** More...
uint8_t	**fingerSearch** (uint8_t slot=1) Ask the sensor to search the current slot fingerprint features to match saved templates. The matching location is stored in **fingerID** and the matching confidence in **confidence** More...
uint8_t	**getTemplateCount** (void) Ask the sensor for the number of templates stored in memory. The number is stored in **templateCount** on success. More...
uint8_t	**setPassword** (uint32_t password) Set the password on the sensor (future communication will require password verification so don't forget it!!!) More...
uint8_t	**LEDcontrol** (bool on) Control the built in LED. More...

uint8_t	**LEDcontrol** (uint8_t control, uint8_t speed, uint8_t coloridx, uint8_t count=0) Control the built in Aura LED (if exists). Check datasheet/manual for different colors and control codes available. More...
uint8_t	**setBaudRate** (uint8_t baudrate) Change UART baudrate. More...
uint8_t	**setSecurityLevel** (uint8_t level) Change security level. More...
uint8_t	**setPacketSize** (uint8_t size) Change packet size. More...
void	**writeStructuredPacket** (const **Adafruit_Fingerprint_Packet** &p) Helper function to process a packet and send it over UART to the sensor. More...
uint8_t	**getStructuredPacket** (**Adafruit_Fingerprint_Packet** *p, uint16_t timeout=**DEFAULTTIMEOUT**) Helper function to receive data over UART from the sensor and process it into a packet. More...

Public Attributes

uint16_t	**fingerID** The matching location that is set by **fingerFastSearch()**
uint16_t	**confidence**
uint16_t	**templateCount** The number of stored templates in the sensor, set by **getTemplateCount()**
uint16_t	**status_reg** = 0x0 The status register (set by getParameters)
uint16_t	**system_id** = 0x0 The system identifier (set by getParameters)
uint16_t	**capacity** = 64 The fingerprint capacity (set by getParameters)
uint16_t	**security_level** = 0 The security level (set by getParameters)
uint32_t	**device_addr** The device address (set by getParameters) More...
uint16_t	**packet_len** = 64 The max packet length (set by getParameters)
uint16_t	**baud_rate** = 57600 The UART baud rate (set by getParameters)

Detailed Description

! Helper class to communicate with and keep state for fingerprint sensors

Constructor & Destructor Documentation

◆ Adafruit_Fingerprint() [1/2]

```
Adafruit_Fingerprint::Adafruit_Fingerprint ( HardwareSerial *  hs,
                                             uint32_t          password = 0x0
                                           )
```

Instantiates sensor with Hardware Serial.

Parameters

- **hs** Pointer to HardwareSerial object
- **password** 32-bit integer password (default is 0)

◆ Adafruit_Fingerprint() [2/2]

```
Adafruit_Fingerprint::Adafruit_Fingerprint ( Stream *  serial,
                                             uint32_t  password = 0x0
                                           )
```

Instantiates sensor with a stream for Serial.

Parameters

- **serial** Pointer to a Stream object
- **password** 32-bit integer password (default is 0)

Member Function Documentation

◆ begin()

```
void Adafruit_Fingerprint::begin ( uint32_t  baudrate )
```

Initializes serial interface and baud rate.

Parameters

- **baudrate** Sensor's UART baud rate (usually 57600, 9600 or 115200)

◆ verifyPassword()

```
boolean Adafruit_Fingerprint::verifyPassword ( void  )
```

Verifies the sensors' access password (default password is 0x0000000). A good way to also check if the sensors is active and responding.

Returns

True if password is correct

◆ getParameters()

uint8_t Adafruit_Fingerprint::getParameters (void)

Get the sensors parameters, fills in the member variables status_reg, system_id, capacity, security_level, device_addr, packet_len and baud_rate.

Returns

True if password is correct

◆ getImage()

uint8_t Adafruit_Fingerprint::getImage (void)

Ask the sensor to take an image of the finger pressed on surface.

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_NOFINGER` if no finger detected

`FINGERPRINT_PACKETRECIEVEERR` on communication error

`FINGERPRINT_IMAGEFAIL` on imaging error

◆ image2Tz()

uint8_t Adafruit_Fingerprint::image2Tz (uint8_t slot = 1)

Ask the sensor to convert image to feature template.

Parameters

slot Location to place feature template (put one in 1 and another in 2 for verification to create model)

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_IMAGEMESS` if image is too messy

`FINGERPRINT_PACKETRECIEVEERR` on communication error

`FINGERPRINT_FEATUREFAIL` on failure to identify fingerprint features

`FINGERPRINT_INVALIDIMAGE` on failure to identify fingerprint features

◆ createModel()

uint8_t Adafruit_Fingerprint::createModel (void)

Ask the sensor to take two print feature template and create a model.

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_PACKETRECIEVEERR` on communication error

`FINGERPRINT_ENROLLMISMATCH` on mismatch of fingerprints

◆ emptyDatabase()

uint8_t Adafruit_Fingerprint::emptyDatabase (void)

Ask the sensor to delete ALL models in memory.

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_BADLOCATION` if the location is invalid

`FINGERPRINT_FLASHERR` if the model couldn't be written to flash memory

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ storeModel()

uint8_t Adafruit_Fingerprint::storeModel (uint16_t location)

Ask the sensor to store the calculated model for later matching.

Parameters

location The model location #

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_BADLOCATION` if the location is invalid

`FINGERPRINT_FLASHERR` if the model couldn't be written to flash memory

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ loadModel()

uint8_t Adafruit_Fingerprint::loadModel (uint16_t location)

Ask the sensor to load a fingerprint model from flash into buffer 1.

Parameters

location The model location #

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_BADLOCATION` if the location is invalid

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ getModel()

uint8_t Adafruit_Fingerprint::getModel (void)

Ask the sensor to transfer 256-byte fingerprint template from the buffer to the UART.

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ deleteModel()

uint8_t Adafruit_Fingerprint::deleteModel (uint16_t location)

Ask the sensor to delete a model in memory.

Parameters

location The model location #

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_BADLOCATION` if the location is invalid

`FINGERPRINT_FLASHERR` if the model couldn't be written to flash memory

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ fingerFastSearch()

uint8_t Adafruit_Fingerprint::fingerFastSearch (void)

Ask the sensor to search the current slot 1 fingerprint features to match saved templates. The matching location is stored in **fingerID** and the matching confidence in **confidence**

Returns

`FINGERPRINT_OK` on fingerprint match success

`FINGERPRINT_NOTFOUND` no match made

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ fingerSearch()

uint8_t Adafruit_Fingerprint::fingerSearch (uint8_t slot = 1)

Ask the sensor to search the current slot fingerprint features to match saved templates. The matching location is stored in **fingerID** and the matching confidence in **confidence**

Parameters

slot The slot to use for the print search, defaults to 1

Returns

`FINGERPRINT_OK` on fingerprint match success

`FINGERPRINT_NOTFOUND` no match made

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ getTemplateCount()

uint8_t Adafruit_Fingerprint::getTemplateCount (void)

Ask the sensor for the number of templates stored in memory. The number is stored in **templateCount** on success.

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ setPassword()

uint8_t Adafruit_Fingerprint::setPassword (uint32_t password)

Set the password on the sensor (future communication will require password verification so don't forget it!!!)

Parameters

password 32-bit password code

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ LEDcontrol() [1/2]

uint8_t Adafruit_Fingerprint::LEDcontrol (bool on)

Control the built in LED.

Parameters

on True if you want LED on, False to turn LED off

Returns

`FINGERPRINT_OK` on success

◆ LEDcontrol() [2/2]

```
uint8_t Adafruit_Fingerprint::LEDcontrol ( uint8_t  control,
                                           uint8_t  speed,
                                           uint8_t  coloridx,
                                           uint8_t  count = 0
                                          )
```

Control the built in Aura LED (if exists). Check datasheet/manual for different colors and control codes available.

Parameters

- **control** The control code (e.g. breathing, full on)
- **speed** How fast to go through the breathing/blinking cycles
- **coloridx** What color to light the indicator
- **count** How many repeats of blinks/breathing cycles

Returns

- `FINGERPRINT_OK` on fingerprint match success
- `FINGERPRINT_NOTFOUND` no match made
- `FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ setBaudRate()

```
uint8_t Adafruit_Fingerprint::setBaudRate ( uint8_t  baudrate )
```

Change UART baudrate.

Parameters

- **baudrate** 8-bit Uart baudrate

Returns

- `FINGERPRINT_OK` on success
- `FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ setSecurityLevel()

```
uint8_t Adafruit_Fingerprint::setSecurityLevel ( uint8_t  level )
```

Change security level.

Parameters

- **level** 8-bit security level

Returns

- `FINGERPRINT_OK` on success
- `FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ setPacketSize()

uint8_t Adafruit_Fingerprint::setPacketSize (uint8_t size)

Change packet size.

Parameters

size 8-bit packet size

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_PACKETRECIEVEERR` on communication error

◆ writeStructuredPacket()

void Adafruit_Fingerprint::writeStructuredPacket (const **Adafruit_Fingerprint_Packet** & packet)

Helper function to process a packet and send it over UART to the sensor.

Parameters

packet A structure containing the bytes to transmit

◆ getStructuredPacket()

uint8_t Adafruit_Fingerprint::getStructuredPacket (**Adafruit_Fingerprint_Packet** * packet,
uint16_t timeout = **DEFAULTTIMEOUT**
)

Helper function to receive data over UART from the sensor and process it into a packet.

Parameters

packet A structure containing the bytes received

timeout how many milliseconds we're willing to wait

Returns

`FINGERPRINT_OK` on success

`FINGERPRINT_TIMEOUT` or `FINGERPRINT_BADPACKET` on failure

Member Data Documentation

◆ confidence

uint16_t Adafruit_Fingerprint::confidence

The confidence of the **fingerFastSearch()** match, higher numbers are more confidents

◆ device_addr

uint32_t Adafruit_Fingerprint::device_addr

Initial value:

```
=
        0xFFFFFFFF
```

The device address (set by getParameters)

The documentation for this class was generated from the following files:

- **Adafruit_Fingerprint.h**
- **Adafruit_Fingerprint.cpp**